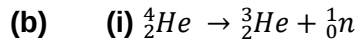


**6(a)** Binding energy is defined as the amount of energy needed to split a nucleus into its individual nucleons.



**(ii)** Q = Difference in total BE [CONCEPT]

$$= 4(6.8465) - 3(2.2666)$$

$$= 20.5862 \text{ MeV}$$

**(iii)** mass of neutron + mass of  ${}^3_2\text{He}$  – mass of  ${}^4_2\text{He} = \frac{Q}{931.494}$

$$\text{mass of } {}^4_2\text{He} - \text{mass of } {}^3_2\text{He} = \text{mass of neutron} - \frac{Q}{931.494}$$

$$= 1.0097 \text{ u} - \frac{20.5862}{931.494}$$

$$= 0.9876 \text{ u}$$

**(iv)** Since energy is supplied to make the process happen, this is a mass excess reaction.

$$\text{So } m_{\text{helium-4}} < m_{\text{helium-3}} + m_{\text{neutron}}$$

$$m_{\text{helium-4}} - m_{\text{helium-3}} < m_{\text{neutron}}$$

**7 (a)** Newton's law of gravitation states that the force of attraction between two point masses is